

A brief overview of Deep Unfolded Fractional Programming-Based Beamforming in RIS-Aided MISO Systems

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I. SYSTEM MODEL

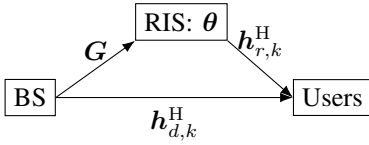


Fig. 1. A base concept of the system.

The Model is described by a Downlink MISO System. It consists of three main elements, the Base Station (BS), the users and the Reconfigurable intelligent surface (RIS). The system has the following components:

- BS has M antennas
- There are K users to be serviced
- RIS has N elements, which causes a phase shift of $\theta \in \mathbb{C}^N$, in each element respectively
- $\mathbf{h}_{d,k} \in \mathbb{C}^M$ is the channel matrix between the BS and the user k .
- $\mathbf{G} \in \mathbb{C}^{N \times M}$ is the channel matrix between the BS and the RIS.
- $\mathbf{h}_{r,k} \in \mathbb{C}^N$ is the channel matrix between the BS and the user k .

Therefore the user k receives the signal:

$$y_k = (\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G}) \sum_{k=1}^K \mathbf{v}_k s_k + n_k \quad (1)$$

from the BS. Where:

- s_k is data symbol for user k
- n_k add. white Gaussian noise. $n_k \sim \mathcal{CN}(0, \sigma^2)$
- $\sum_{k=1}^K \mathbf{v}_k s_k = \mathbf{x}$ the signal send by the the BS. ?
- $\mathbf{v}_k \in \mathbb{C}^{M \times 1}$ is the beamforming vector from the BS to the user k .
- $\mathbf{h}_{d,k}^H$ is direct path between the BS and user k .
- $\mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G}$ is the path from the BS to RIS, application of the phase shift and then the path form the RIS to the user k .
- $\text{diag}(\theta)$ simply creates a diagonal matrix with the vector.
- Summary: $y_k = (\mathbf{h}_{dir}^H + \mathbf{h}_{indir}^H) \mathbf{x} + n_k$

The SINR can be calculated as:

$$\alpha_k = \frac{\left| \left(\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G} \mathbf{v}_k \right) \right|^2}{\sum_{i=1, i \neq k}^K \left| \left(\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G} \mathbf{v}_i \right) \right|^2 + \sigma_0^2} \quad (2)$$

which is simple the expected signal divided by the sum of the noise and all other signals send.

The objective is to maximize the weighted sum rate (WSR):

$$(\mathcal{P}_1) : \max_{\mathbf{V}, \theta} \sum_{k=1}^K \omega_k \log(1 + \alpha_k) \quad (3)$$

$$\text{s.t. } |\theta_n| = 1, \forall n, \quad (4)$$

$$\sum_{k=1}^K \|\mathbf{v}_k\| \leq P_{\max} \quad (5)$$

The system allows changing the beamforming vectors of the BS and the phase shift values of the RIS elements. $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K]$ and ω_k a weight coefficient for the user k .

This is a typical multi-ratio fractional programming problem, with a possibly non-convex solution space.

II. THE DESIGN, INTENT, AND LIMITATIONS OF THE TEMPLATES